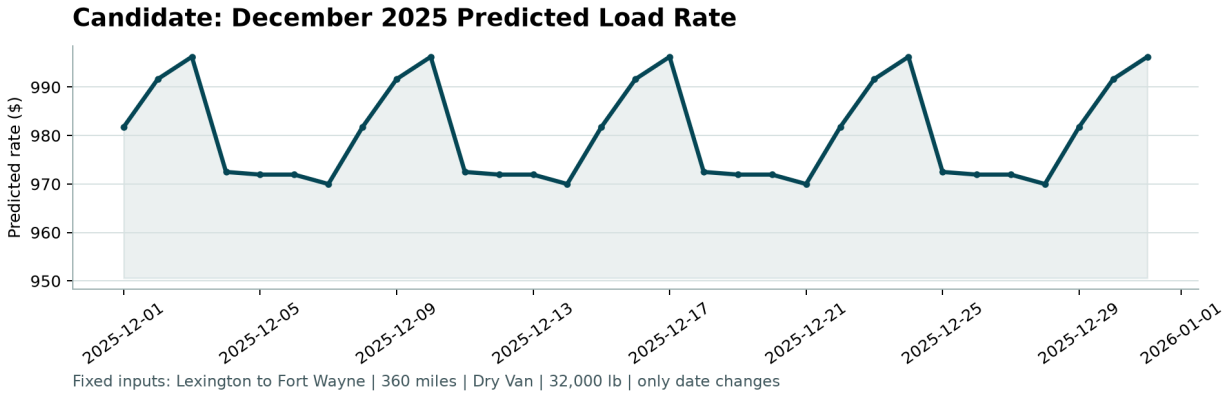
1. Approach to Validate Test/Train Data and Split

- **Chronological Time-Based Split:** To prevent data leakage and simulate real-world production deployment, data was split chronologically rather than randomly. Past historical data was used to train the model, while out-of-time future periods formed the test set.
- **Chronological Rolling-Window Validation:** To simulate real-world production deployment and prevent look-ahead bias, validation was structured using a rolling monthly time split (training on past rolling months and validating on the subsequent out-of-time month).

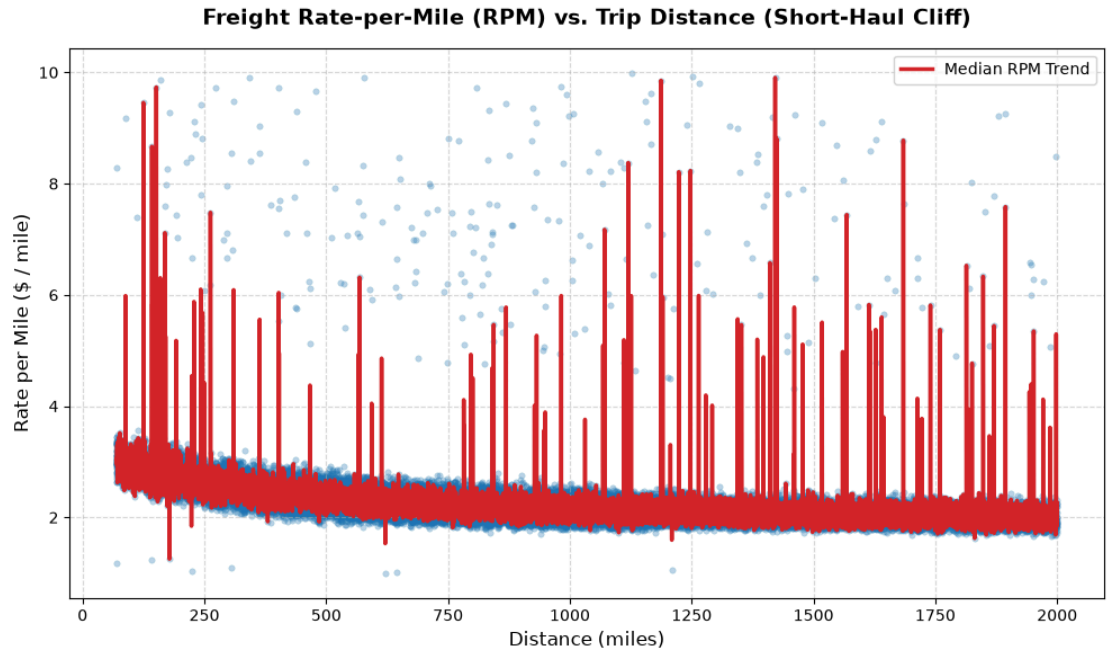
2. December 2025 Predicted Load Rate



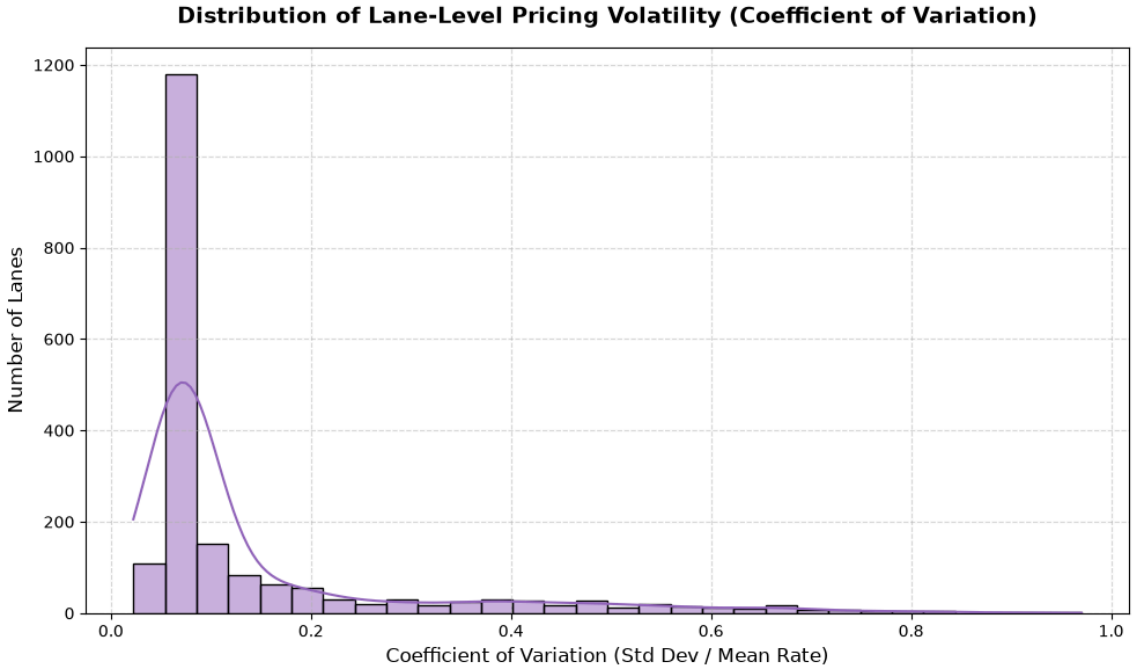
- **Forecast Behavior:** The model successfully captures weekly operational cycles and rate floors (~970) with predictable peaks (990+), reflecting weekday demand spikes and weekend stabilization in industrial freight lanes.

Part 1: Key Findings from Data Exploration (EDA)

- **Short-Haul Cliff (RPM vs. Distance):** Rate-per-mile spikes dramatically for short trips (under 500 miles) due to fixed loading/unloading overheads, then flattens out into a stable decay curve for long hauls.



- **Lane Pricing Volatility:** The distribution of lane volatility shows a heavy peak near zero, proving that the vast majority of freight corridors have highly stable pricing week-over-week, with only a small long-tail of erratic lanes.



- **Weight & Equipment Yields:** Scatter plots demonstrate that heavy loads do not automatically inflate rates across the board; baseline pricing is driven primarily by lane distance and equipment constraints rather than raw weight alone.



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Part 2: Data Quality Issues & How They Were Addressed

- **Negative Weights:** Detected negative weight entries, verified their absolute values fell within standard equipment bounds, and safely converted them to positive values across training and inference pipelines.

- **Missing Weights:** Imputed missing weight values using the median weight of the specific equipment type, falling back to the global median if necessary.

- **Missing Market Index:** Resolved missing macroeconomic index values using a hierarchical fallback strategy:
    1. Median of the same route and month combination.
    2. Fallback to route-level median if the route-month combo was missing.
    3. Final safety fallback to the global median.

- **Lane Outliers & Anomalies:** Cleaned pricing anomalies by computing a robust outlier score based on Median Absolute Deviation (MAD) from the lane median. Rows with an outlier score ≥ 6.5 had their `posted_rate` smoothed to the lane median.
    1. **Find the Baseline:** We calculated the typical median price for every specific shipping route.
    2. **Measure Deviation:** We checked how far each trip strayed from that route's normal price using a robust statistical yardstick (Median Absolute Deviation).
    3. **Flag Extreme Errors:** Any price landing **6.5 times further away** than normal fluctuations allowed was flagged as a data entry typo.
    4. **Safe Smoothing:** We replaced those rare extreme errors with the route's safe median so our AI model isn't trained on bad data.

- **Spatial Validation:** Computed Haversine distances from coordinates and verified a near-perfect correlation (~ 0.999) with the dataset's `distance` column, confirming high spatial data integrity.

Part 3 & 4: Model Reasoning, Training, and Validation Approach

- **Model Selection:** We benchmarked 5 architectures—Linear Regression, Linear SVR, Random Forest, XGBoost, and LightGBM—using a rigorous **rolling-month cross-validation** scheme (`rolling_month_cv`) with a 5-month training window. Gradient boosted trees (specifically LightGBM) won out because they natively capture non-linear feature interactions (like distance decay and weight thresholds) while maintaining sub-100ms inference speeds.

- **Train/Val Split Strategy:** To prevent future data leakage and mirror real-world production conditions, the final model was trained on historical months except last month and strictly evaluated on out-of-time data from the absolute final month.

Part 5: Code Walkthrough (Key Components)

- **1. EDA & Feature Extraction:** Explored multi-dimensional freight patterns (short-haul distance decay, lane pricing stability and equipment weight yields) to inform feature engineering and modeling constraints.

- **2. Data Cleaning & Robust Imputation:** Handled anomalies and data errors by cleaning negative weights, removing lane outliers using Median Absolute Deviation (MAD), and applying a hierarchical fill strategy (route-month, route, and global medians) for missing weights and market indices.

- **3. Time-Based Cross-Validation & Model Selection:** Benchmarked models using a chronological rolling-month validation strategy (training on a rolling 5-month window and validating on the subsequent month), ultimately selecting LightGBM as the winning model based on superior RMSE and MAE performance.

- **4. Full Model Training & Artifact Bundling:** Trained the final LightGBM model on historical data and serialized both the model and critical imputation statistics (equipment weight medians, route-month market baselines) into a portable `model_bundle.pkl` file using `joblib` to ensure complete training-inference parity.

- **5. Production Refactoring & Inference Pipeline:** Built a modular `score.py` pipeline (`preprocess_and_impute_inference`) that loads the saved bundle artifact to cleanly preprocess, impute, and generate batch or single-load forward predictions (such as the December chart inputs) for final submission.